Rubik's Cube Solver

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Introduction

- The 3 x 3 x 3 Rubik's Cube, invented by Ernő Rubik in 1974, is a multidimensional puzzle. The objective is to align all squares of the same color on each side after the cube has been scrambled.
- The Rubik's Cube solver project aims to build an autonomous robotic system and a real-time graphical user interface.

Specifications and Design Requirements

- **Robotic System:** A system consisting of 6 independent NEMA 17 stepper motors is built to solve a 3 x 3 x 3 Rubik's Cube
- **Performance:** The objective is to achieve a solution in less than 20 moves using Kociemba's algorithm.
- **Electronics:** A custom PCB is utilized to manage motor driver synchronization and power distribution.
- **User Interface:** A real-time graphical interface is developed to display the live cube state and provide manual correction tools.

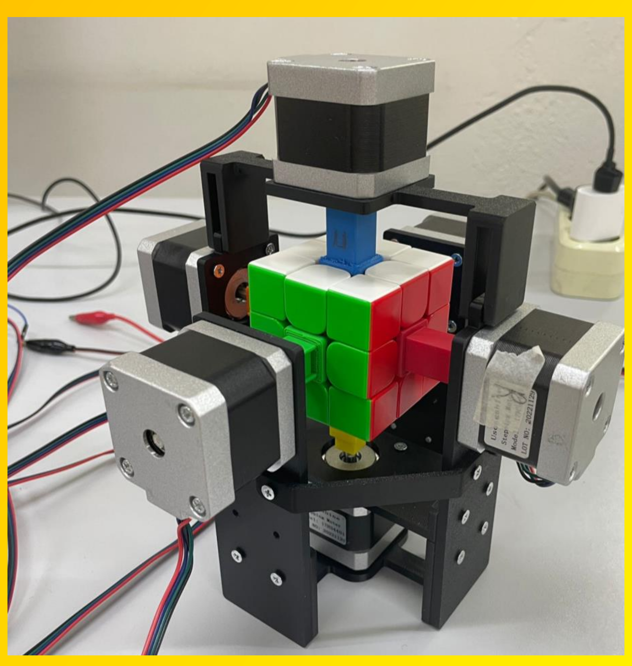

Figure 1: Mechanical System

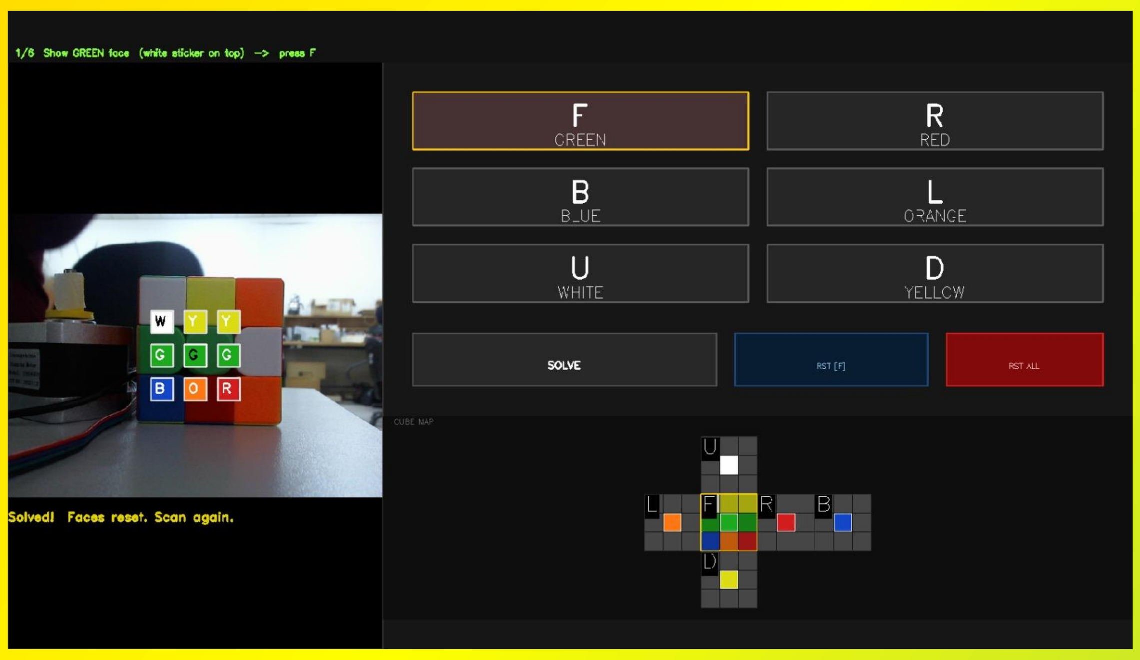

Figure 2: Custom Graphical User Interface (GUI)

Solution Methodology

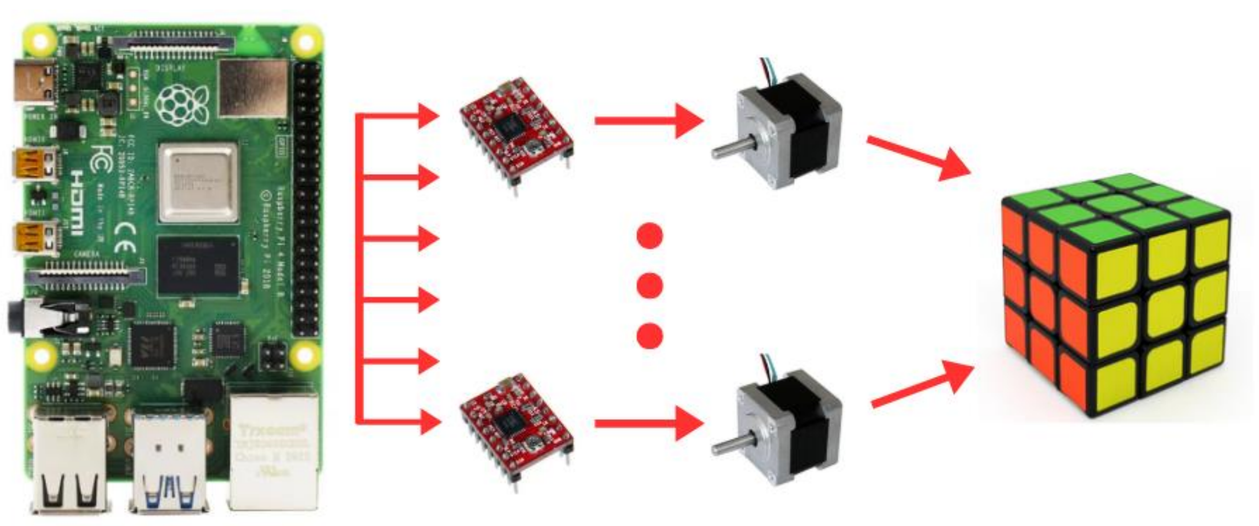

Figure 3 : Electromechanical architecture

- **Central Controller:** Raspberry Pi 4b coordinates computer vision and motor signals.
- **Motor Control:** 6 A4988 drivers control 6 NEMA 17 motors via GPIO pins.
- **Power Management:** A custom PCB centralizes connections and ensures stable voltage distribution.

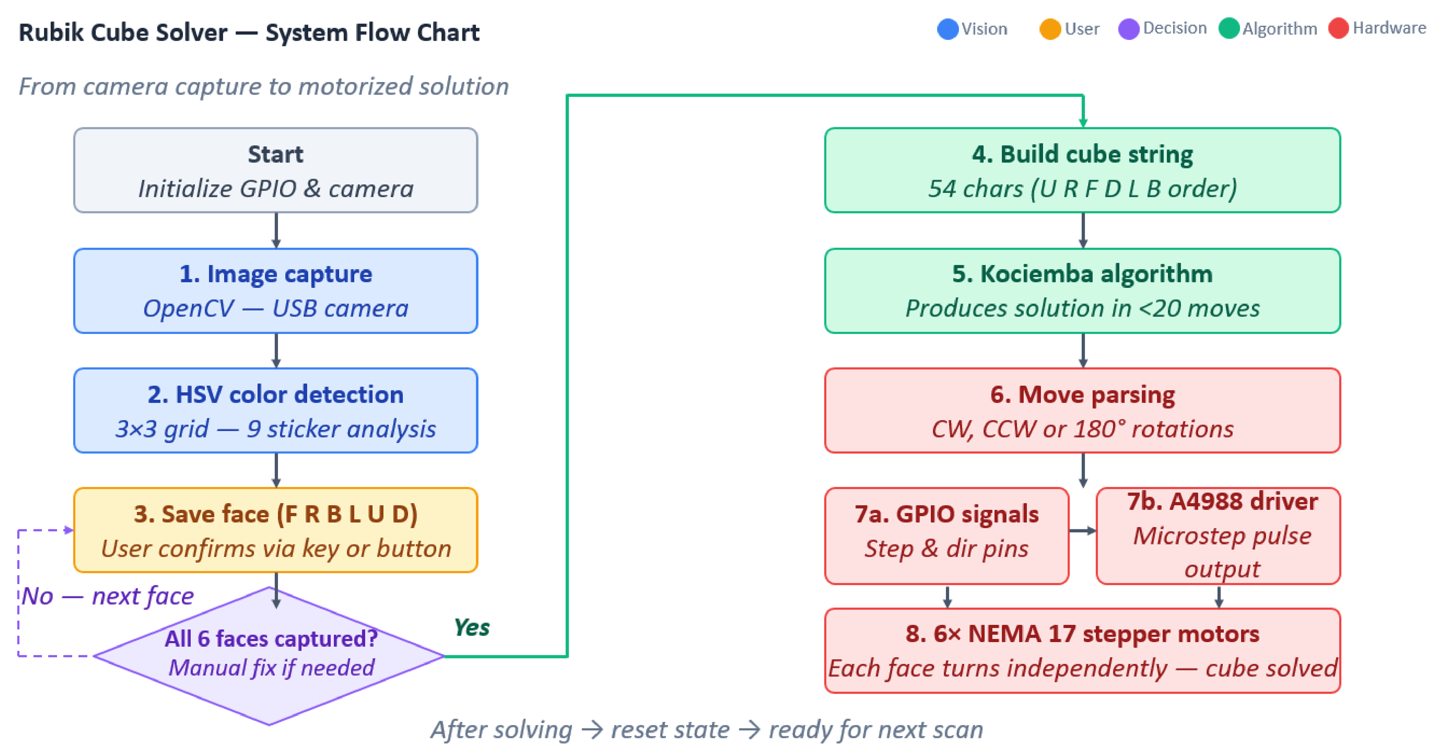

Figure 4 : State Flowchart

Application Areas

- **Education:** Hands on platform for robotics, computer vision, and embedded systems courses.
- **Research:** Low cost testbed for benchmarking color detection and motion planning algorithms.
- **Entertainment:** Demonstrates autonomous puzzle solving at science fairs and public exhibitions.
- **Industry:** Multi-motor control architecture is transferable to pick and place and automated sorting systems.

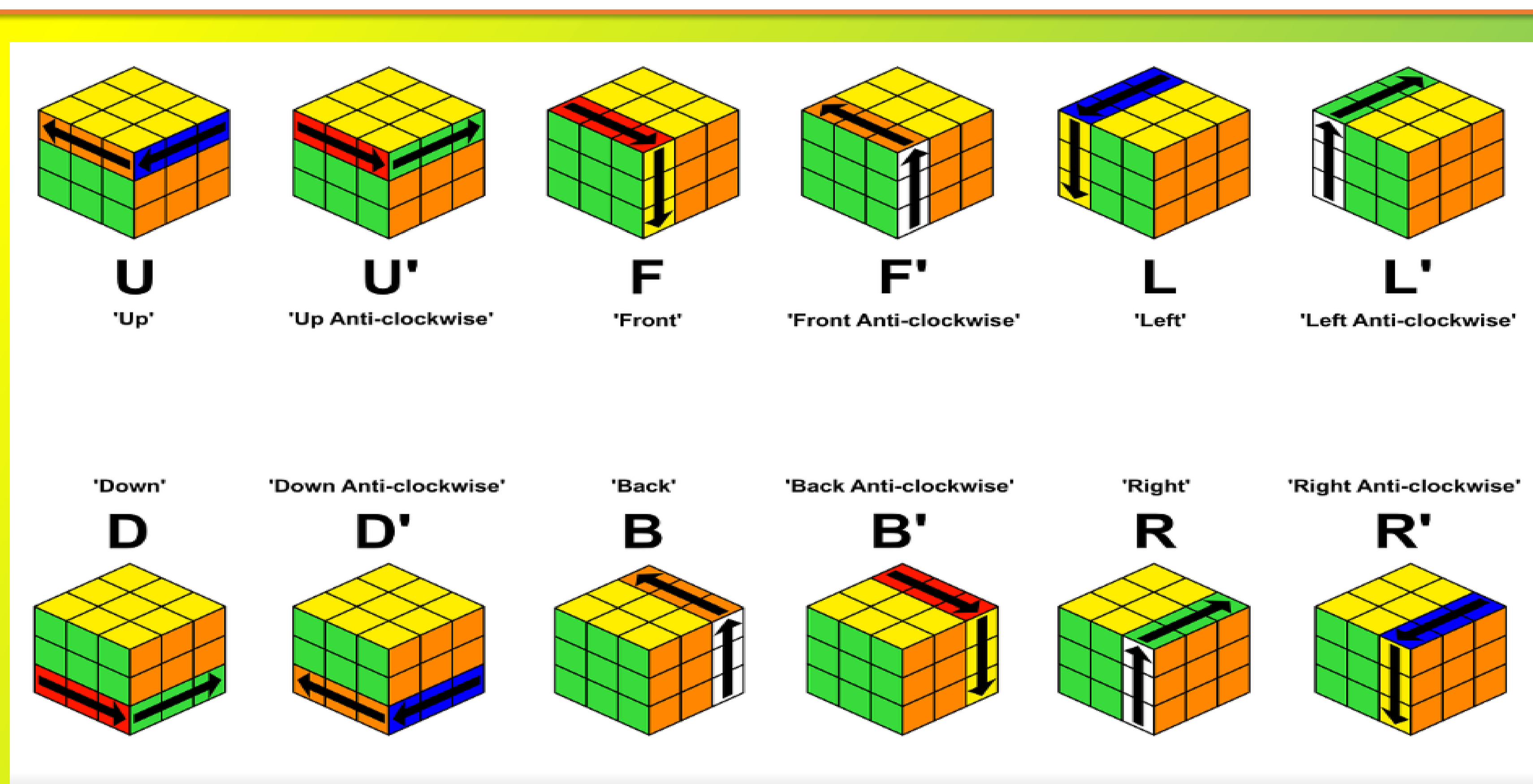

Figure 5: Standard Rubik's Cube move notation

Results and Discussion

- **Results:** The system successfully solves a 3 x 3 x 3 Rubik's Cube autonomously using 6 NEMA 17 stepper motors and a Raspberry Pi 4B. HSV-based color detection identifies all 6 face colors via webcam, supported by a custom GUI for manual verification.
- **Performance:** The system achieves a high-speed solve, Kociemba's algorithm generates optimal solutions (under 20 moves) in.
- **Specifications Achieved:** 6-motor high-speed synchronized control via a custom PCB, real-time OpenCV color detection, and a functional graphical interface were successfully implemented.
- **Implications:** The integration of high speed actuation and real-time computer vision demonstrates an advanced level of system coordination and motion planning.
- **Future Work:** Adaptive lighting compensation and closed loop feedback could further increase color detection robustness and rotation precision.

References

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